

STAR PNT INTERNSHIP REPORT
ON
Detection & Classification of Spoofing Attacks on GNSS Signals
Using AI-Based Models on NavIC System

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CERTIFICATE

This is to certify that the internship report titled DETECTION & CLASSIFICATION OF SPOOFING ATTACKS ON GNSS SIGNALS USING AI-BASED MODELS ON NAVIC SYSTEM submitted by YUNES ABDULGHANI OHAMMED GHALEB Department of Computer Science and IT (CSIT), JAIN (Deemed-to-be University), is a Bonafide record of the original work carried out by the student during the STAR PNT Internship Program conducted under the Positioning, Navigation and Timing (PNT) Laboratory, supported by the IIT Tirupati Navavishkar I-Hub Foundation (IITTNiF).

The internship work was successfully completed under the coordination and supervision of Dr. K. Suneetha, Professor & Head, Department of CSIT and Dr. Zion R., Assistant Professor, Department of CSIT, JAIN (Deemed-to-be University).

This certificate is issued as part of the final evaluation and review process of the internship program.

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(Signature)	(Signature)

Internal Examiner(s)	External Examiner(s)
1.	1.
2.	2.

DECLARATION

I hereby declare that the internship report titled DETECTION & CLASSIFICATION OF SPOOFING ATTACKS ON GNSS SIGNALS USING AI-BASED MODELS ON NAVIC , submitted to the Positioning, Navigation and Timing (PNT) Laboratory, under the STAR PNT Internship Program, completed under JAIN (Deemed-to-be University) Nodal Centre and supported by the IIT Tirupati Navavishkar I-Hub Foundation (IITTNiF), is a Bonafide record of the original work carried out by me during the internship period.

This report is the result of my independent project work completed under the valuable coordination and academic support of my Project Coordinators Dr. K. Suneetha, Professor & Head, Department of CSIT, and Dr. Zion, Assistant Professor, Department of CSIT, JAIN (Deemed-to-be University). Their expert guidance, constructive feedback, and constant encouragement have been instrumental in the successful completion of my internship project.

I firmly declare that this report has not been submitted to any other institution, organization, or university for the award of any degree, diploma, or certificate. All the information, data, figures, and analysis presented in this report are based on my original work carried out during the internship. Any content or material drawn from other sources has been duly acknowledged and cited in the References section.

I take full responsibility for the authenticity, accuracy, and originality of the work presented in this report and affirm that it truly reflects my learning outcomes and practical experience gained during the STAR PNT Internship.

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Date: 01/11/2025

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Date: 30th October

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List of Abbreviations

<u>Abbreviation</u>	<u>Full Form / Meaning</u>
AI	Artificial Intelligence
DNN	Deep Neural Network
GNSS	Global Navigation Satellite System
NavIC	Navigation with Indian Constellation
RF	Random Forest
SVM	Support Vector Machine
XGBoost	Extreme Gradient Boosting
PNT	Position, Navigation, Timing
INS	Inertial Navigation System
C/N0	Carrier-to-Noise Ratio
SNR	Signal-to-Noise Ratio
FN	False Negative
FP	False Positive
TP	True Positive
TN	True Negative
ROC	Receiver Operating Characteristic
AUC	Area Under Curve
PR	Precision-Recall
AP	Average Precision
CV	Cross Validation

K-Fold	K-Fold Cross Validation
ETA	Learning Rate (in XGBoost)
Subsample	Row Sampling Ratio for Boosting
Colsample_bytree	Column Sampling Ratio per Tree
Scaler	Feature Normalization Object
Label	Target Variable (Normal / Spoofing)
ROC-AUC	Receiver Operating Characteristic - Area Under Curve
PR-AUC	Precision Recall - Area Under Curve
FP Rate	False Positive Rate
TP Rate	True Positive Rate
LogLoss	Logarithmic Loss Function
ACC	Accuracy
STD	Standard Deviation
R ²	Coefficient of Determination
CV Results	Cross Validation Performance Summary
Booster Predict	Prediction Function for Trained Model
Eval Metric	Metric Evaluation Used During Training
Confusion Matrix	Table of True/False Predictions
Precision	$TP / (TP + FP)$
Recall	$TP / (TP + FN)$
F1-Score	Harmonic Mean of Precision and Recall
True Label	Actual Class of Sample

Predicted Label	Model Output Class
Spoofing	Falsified GNSS Signal
Normal	Authentic GNSS Signal
GPS	Global Positioning System
GLONASS	Global Navigation Satellite System (Russia)
Galileo	European GNSS System
BeiDou	Chinese Navigation Satellite System

Abstract

Global Navigation Satellite Systems (GNSS) play an indispensable role in providing precise Positioning, Navigation, and Timing (PNT) information essential for modern transportation, defense, and communication infrastructures. Despite their global significance, GNSS signals remain inherently vulnerable to spoofing attacks malicious attempts to transmit counterfeit satellite signals that mislead receivers and compromise navigation integrity. As the dependency on GNSS, particularly India's Navigation with Indian Constellation (NavIC), continues to grow, developing intelligent, adaptive, and data-driven spoofing detection mechanisms has become a critical security priority.

This research presents a comprehensive comparative study of three state-of-the-art machine learning models—Deep Neural Network (DNN), Random Forest (RF), and Extreme Gradient Boosting (XGBoost)—for detecting spoofing attacks in real-world GNSS signals. The models were trained on a dataset comprising 74,084 labeled samples and validated on an independent test set of 36,189 samples. Rigorous preprocessing was applied, including feature engineering (Doppler shift, carrier-to-noise ratio (C/N_0)), normalization, and class balancing to ensure fair and robust model performance evaluation.

Experimental results demonstrate that all three models exhibit strong classification performance; however, XGBoost outperformed the others, achieving the highest accuracy of 95.5%, a recall of 95.4%, and minimizing False Negatives (FN) to just 334 cases, thereby establishing its superiority for mission-critical GNSS security applications. The improved DNN also exhibited excellent generalization, successfully classifying 36,188 out of 36,189 unseen samples, reinforcing its reliability for dynamic real-world environments.

This study contributes a novel empirical validation of AI-based GNSS spoofing detection using real NavIC data, bridging a crucial research gap in regional satellite navigation security. Unlike earlier works that rely on simulated datasets, this research validates model robustness in authentic conditions and establishes a benchmark framework for future AI-based GNSS protection systems. The findings underscore that AI-driven boosting algorithms—particularly XGBoost—offer an optimal balance between interpretability, computational efficiency, and accuracy, paving the way for secure, autonomous, and resilient GNSS architectures.

Chapter.1

Introduction

1.1 Background

Global Navigation Satellite Systems (GNSS) have become one of the most essential technological backbones of modern civilization. They provide the critical foundation for Positioning, Navigation, and Timing (PNT) services, supporting applications in transportation, aviation, telecommunications, agriculture, disaster management, and defense. Over the past decades, multiple global and regional navigation systems have been developed, including the Global Positioning System (GPS) of the United States, GLONASS of Russia, Galileo of the European Union, BeiDou of China, and India's regional system, NavIC (Navigation with Indian Constellation). These constellations collectively ensure global coverage, providing centimeter-level positioning accuracy to billions of users daily.

As the dependence on GNSS-based technologies continues to grow, so does the vulnerability of these systems to intentional interference. Among the most critical forms of interference is spoofing, wherein adversaries transmit counterfeit satellite signals to mislead a receiver into calculating an incorrect position, velocity, or timing. Spoofing attacks can have devastating consequences, from misleading unmanned aerial vehicles (UAVs) and autonomous cars to disrupting financial transactions, telecommunications synchronization, and defense operations. Therefore, securing GNSS infrastructure has become a strategic priority in both civilian and military contexts.

1.2 Problem Definition

Spoofing attacks have evolved from simple signal imitation to sophisticated, multi-channel, time-synchronized forgeries capable of deceiving even advanced GNSS receivers. Modern spoofers can replicate authentic signal structures, modulations, and Doppler characteristics, making traditional detection methods increasingly ineffective. While conventional defense strategies—such as signal power monitoring, correlation peak distortion analysis, and antenna-based angle-of-arrival estimation—offer partial protection, they often require specialized hardware and perform poorly under dynamic conditions.

The core problem lies in the inability of these traditional techniques to adapt dynamically to new spoofing patterns and minimize false negatives (FN)—cases where spoofed signals are incorrectly identified as authentic. False negatives represent the most dangerous type of error, as they can lead to undetected navigation compromises with potentially catastrophic outcomes.

Given the growing sophistication of spoofing strategies, there is an urgent need for data-driven, adaptive detection mechanisms capable of recognizing complex patterns and generalizing effectively to unseen attack scenarios using real-world GNSS data.

1.3 Motivation and Research Gap

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have transformed various domains, offering models capable of uncovering nonlinear patterns hidden in high-dimensional data. In the field of GNSS security, AI-based approaches show tremendous potential for learning spoofing signatures from signal behavior — such as Doppler shift, carrier-to-noise ratio (C/N_0), and pseudo-range variations — without relying on physical hardware modifications.

However, most prior studies on GNSS spoofing detection have relied on simulated or synthetic datasets, which fail to capture the full variability and noise characteristics of real satellite signals. Furthermore, limited research has focused specifically on NavIC, India's regional navigation system, despite its growing strategic importance and regional adoption. This creates a critical research gap: there is a lack of comprehensive, comparative, and experimentally validated AI frameworks trained on real-world GNSS/NavIC signal data.

Motivated by this gap, the present study seeks to develop and rigorously evaluate multiple AI-based spoofing detection frameworks using authentic signal datasets, establishing a robust foundation for real-time GNSS security solutions.

1.4 Research Objectives

The overarching goal of this research is to design, implement, and evaluate a high-performance AI-based framework for spoofing detection in GNSS signals, focusing on real-world NavIC and GPS data. The specific objectives are:

1. To acquire and preprocess authentic GNSS signal data, ensuring balanced and high-quality datasets through feature normalization and outlier mitigation.
2. To develop and compare three advanced machine learning models — Deep Neural Network (DNN), Random Forest (RF), and Extreme Gradient Boosting (XGBoost) — under identical experimental settings.
3. To evaluate model performance using comprehensive metrics including accuracy, precision, recall, F1-score, and confusion matrices.
4. To minimize False Negatives (FN) the most critical metric for GNSS spoofing detection and identify the most reliable model for deployment.
5. To explore real-time integration potential of the optimal AI model into UAV and autonomous navigation systems for practical spoofing defense.

1.5 Scope and Contributions

This research focuses primarily on the signal-processing and algorithmic dimensions of GNSS spoofing detection. Physical-layer mitigation, cryptographic authentication, or hardware-based defenses are beyond the immediate scope of this study. Instead, the emphasis is placed on AI-based feature-driven classification, where learning algorithms discern authentic from spoofed signals using only derived signal parameters.

The major contributions of this work are as follows:

- Development of a complete data processing and feature engineering pipeline for GNSS/NavIC signals, including Doppler shift, C/N₀ variations, and temporal stability metrics.
- Comparative performance evaluation of DNN, RF, and XGBoost models on a large-scale real-world dataset containing over 110,000 labeled samples.

2.Problem Statement

In recent years, the global dependence on Global Navigation Satellite Systems (GNSS) has increased dramatically, making them a critical part of modern infrastructure. Systems such as GPS (United States), GLONASS (Russia), Galileo (European Union), BeiDou (China), and India's regional system NavIC (Navigation with Indian Constellation) provide essential positioning, navigation, and timing (PNT) services. These systems support numerous civilian and military applications, including aviation, maritime navigation, precision agriculture, telecommunications, disaster management, and defense operations.

However, this widespread reliance has also made GNSS highly vulnerable to intentional signal manipulation, particularly spoofing attacks, where an adversary transmits counterfeit satellite-like signals to mislead receivers into estimating false positions, velocities, or timing information.

Spoofing represents one of the most severe security threats to GNSS systems because it can deceive receivers without immediate detection. Modern spoofers can replicate the structure, frequency, and timing of genuine satellite signals, and even perform *seamless takeovers* by gradually overpowering authentic signals. As a result, a receiver under spoofing attack may continue to operate normally while producing completely erroneous navigation data. The potential consequences of undetected spoofing are critical ranging from disrupted flight operations and maritime mis navigation to compromised UAV missions and military system failures. This underscores the urgent need for reliable, adaptive, and real-time spoofing detection mechanisms.

Although a variety of traditional spoofing detection techniques have been proposed, each suffers from significant limitations. Signal power monitoring and correlation peak analysis can identify anomalies in controlled environments, but they often fail under complex or dynamic conditions. Multi-antenna and angle-of-arrival methods can be effective but require additional hardware and are unsuitable for small or cost-sensitive devices. Moreover, many existing systems rely on fixed threshold-based detection, which introduces a trade-off between sensitivity and reliability:

- High sensitivity may cause excessive False Positives (FP), triggering unnecessary alarms.
- Low sensitivity increases False Negatives (FN), where actual spoofing attacks remain undetected.

Among these, False Negatives pose the greatest danger, as they allow malicious interference to persist unnoticed, potentially leading to mission-critical failures.

The rise of Artificial Intelligence (AI) and Machine Learning (ML) presents new opportunities to address these challenges. AI-based models can automatically learn and identify complex spoofing patterns from features such as Doppler shift, carrier-to-noise ratio (C/N_0), and signal stability, without relying on specialized hardware. Data-driven approaches have shown remarkable success in identifying anomalies and generalizing across unseen attack scenarios. However, most existing research depends heavily on synthetic or simulated GNSS datasets, which do not accurately reflect

real-world noise, multipath effects, or receiver dynamics. This limits the real-world applicability of the proposed models.

Furthermore, very few studies have performed a systematic comparative analysis of different AI classifiers—such as Deep Neural Networks (DNN), Random Forests (RF), and Extreme Gradient Boosting (XGBoost) using authentic GNSS or NavIC datasets. Without such comparative evaluation, it remains unclear which model provides the best balance between detection accuracy, robustness, and computational efficiency for real-time deployment in operational environments. The lack of real-world validation and large-scale testing creates a significant research gap in the field of intelligent GNSS spoofing detection.

Therefore, the core research problem addressed in this study is the design and evaluation of a scalable AI-based spoofing detection framework capable of minimizing False Negatives while maintaining high overall accuracy and computational feasibility. The objective is to identify the most effective machine learning model that achieves an optimal trade-off between detection sensitivity, model generalization, and processing efficiency when applied to real GNSS and NavIC signal data.

A conceptual overview of the GNSS spoofing detection problem is illustrated in Figure 3.1, showing the interaction between authentic satellite signals, injected spoofed signals, and the AI-based classification system designed to distinguish between genuine and deceptive transmissions.

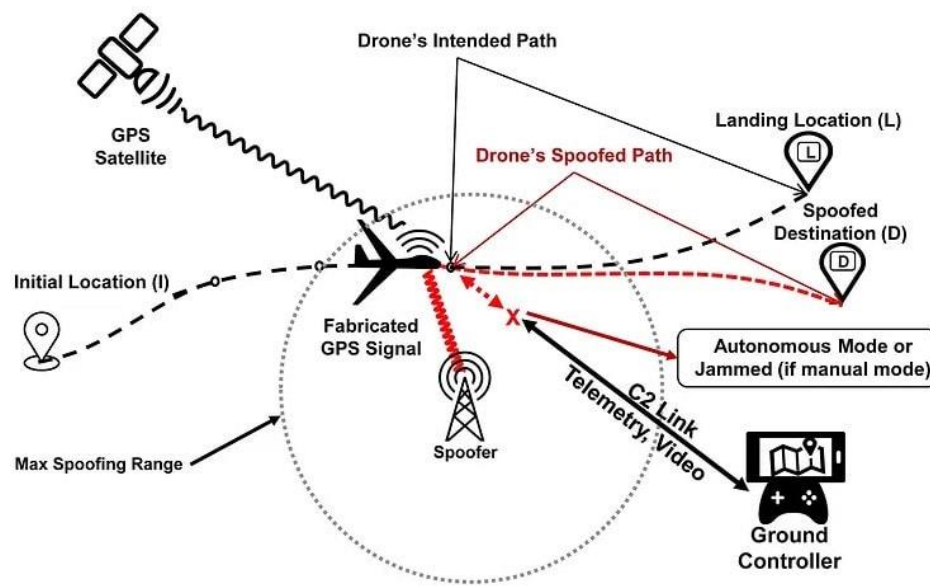


Figure2.1: The Operational Impact of GNSS Spoofing Attack on a UAV's Trajectory, Diverting the Intended Path to a Spoofed Destination.

3.Objectives

The primary objective of this research is to design, develop, and evaluate a robust Artificial Intelligence (AI)–based framework for detecting spoofing attacks in Global Navigation Satellite System (GNSS) signals, particularly focusing on the Navigation with Indian Constellation (NavIC) system. Given the growing threat of signal spoofing and the limitations of traditional detection methods, this study aims to establish a data-driven, scalable, and reliable solution capable of real-time operation under diverse signal conditions.

The first objective is to collect and preprocess real-world GNSS signal data under both authentic and spoofed conditions. This ensures that the dataset used for training accurately reflects real operational environments rather than synthetic data. Preprocessing includes cleaning and standardizing raw data to remove noise, handle missing or corrupted values, and normalize numerical attributes using Standard Scaling techniques. In addition, feature engineering will be applied to extract key characteristics such as Doppler shift, carrier-to-noise ratio (C/N_0), and pseudo-range error, which serve as critical indicators of signal authenticity and stability.

The second objective is to develop and train three machine learning models—a Deep Neural Network (DNN), a Random Forest (RF), and an Extreme Gradient Boosting (XGBoost) classifier—using the curated dataset. Each model represents a distinct learning paradigm: DNN for deep feature abstraction, RF for ensemble-based decision aggregation, and XGBoost for optimized gradient boosting. The models will be tuned and optimized to achieve high accuracy while maintaining computational efficiency suitable for real-time systems. Training will employ cross-validation and early-stopping to ensure generalization and prevent overfitting.

The third objective is to conduct a comparative evaluation of the three AI models using multiple performance metrics—Accuracy, Precision, Recall, F1-Score, and Confusion Matrix—to identify which model demonstrates superior robustness and consistency across unseen test data. Additional metrics such as ROC-AUC and PR-AUC will further assess each model’s capability to distinguish between spoofed and authentic signals.

A central focus of this research is the minimization of False Negatives (FN)—cases where spoofing events go undetected. While overall accuracy is important, the FN rate represents the most critical risk in security-sensitive applications. The goal is to identify the model that minimizes FN while maintaining high detection reliability and low false alarms.

Finally, this study aims to recommend a scalable and deployable AI-based framework for GNSS security systems. The proposed solution emphasizes computational efficiency, interpretability, and adaptability for integration into Unmanned Aerial Vehicle (UAV) navigation platforms or ground-based monitoring stations. By achieving these objectives, the research contributes to enhancing GNSS resilience, improving situational awareness against spoofing threats, and advancing AI-driven navigation security.

Chapter.2

Related Work

Research on securing Global Navigation Satellite Systems (GNSS) against spoofing attacks has significantly intensified in recent years, reflecting the increasing reliance of global infrastructure on precise Positioning, Navigation, and Timing (PNT) services. GNSS systems—including GPS, GLONASS, Galileo, BeiDou, and NavIC—form the foundation for modern transportation, telecommunications, defense, and autonomous systems. However, the open and unencrypted nature of most civilian GNSS signals exposes them to spoofing, where an adversary transmits counterfeit signals to manipulate a receiver's computed position or timing. This vulnerability makes spoofing detection a critical topic in navigation security research.

Early GNSS security studies primarily focused on traditional detection techniques based on signal power, correlation patterns, or physical characteristics. Notable examples include signal power monitoring, correlation peak distortion analysis, and code- and carrier-phase consistency checks. Humphreys et al. (2008) and Psiaki & Humphreys (2016) demonstrated several such techniques, emphasizing how variations in signal power and correlation can expose spoofed signals. While effective under controlled conditions, these methods often fail in complex real-world environments, where multipath effects, dynamic motion, and natural noise obscure spoofing patterns. Moreover, many of these methods require specialized or multi-antenna hardware, limiting their applicability in compact or cost-sensitive receivers.

The introduction of Artificial Intelligence (AI) and Machine Learning (ML) has transformed spoofing detection from rule-based signal analysis into a data-driven classification problem. ML techniques can model nonlinear relationships and detect subtle statistical deviations between authentic and spoofed signals. Several studies have proposed supervised learning models using labeled GNSS datasets. For example, Bian et al. (2019) applied Support Vector Machines (SVMs) to detect spoofing through carrier-to-noise ratio (C/N_0) variations, achieving strong detection accuracy but limited scalability. Similarly, Feng et al. (2020) employed Convolutional Neural Networks (CNNs) to classify GPS L1 signals, demonstrating excellent accuracy but at the cost of heavy computational requirements unsuitable for real-time or embedded applications.

Recent works have shifted toward ensemble-based approaches, such as Random Forest (RF) and Extreme Gradient Boosting (XGBoost), which combine multiple decision trees to achieve both accuracy and robustness. These models outperform earlier classifiers in handling nonlinear, noisy, and high-dimensional GNSS features while remaining computationally efficient. Studies have shown that ensemble learners not only achieve high accuracy but also allow greater interpretability—helping to identify which input features (such as Doppler shift or C/N_0) contribute most to spoofing detection. Furthermore, hybrid frameworks combining deep learning with ensemble models are emerging, aiming to balance feature-learning capability with efficiency.

Despite promising progress, a major limitation across existing AI-based research is the reliance on simulated or laboratory-generated GNSS datasets. Synthetic data, while convenient for model training, fail to capture real-world signal complexity, atmospheric variation, or authentic interference patterns. Consequently, models trained under ideal conditions often exhibit poor generalization when deployed in the field. Only a few studies have utilized real-world GNSS data, and even fewer have performed comparative evaluations of multiple AI models using consistent datasets. This lack of empirical validation leaves open questions regarding which model family—deep learning or ensemble learning—offers the most reliable performance in operational conditions.

In the case of NavIC, India's regional navigation satellite system, research remains comparatively limited. Most existing works focus on interference or signal integrity analysis rather than dedicated AI-based spoofing detection frameworks. While some studies have analyzed NavIC's signal structure, multipath behavior, and anti-jamming capabilities, few have explored machine learning models trained on authentic NavIC signal data. The scarcity of standardized datasets and comparative model analyses restricts the development of secure, adaptive NavIC receivers for defense and autonomous applications.

The present study aims to bridge this research gap by presenting a comprehensive comparative analysis of three state-of-the-art AI models: Deep Neural Network (DNN), Random Forest (RF), and Extreme Gradient Boosting (XGBoost). Unlike prior research relying on simulated signals, this study uses real-world GNSS data containing both genuine and spoofed samples. Rigorous preprocessing—including feature engineering (Doppler shift, C/N_0 , pseudo-range error), standard scaling, and class balancing—ensures high-quality inputs for robust model training. Each model

represents a unique learning paradigm: DNN for deep representation learning, RF for ensemble decision aggregation, and XGBoost for optimized gradient-based boosting.

The study evaluates model performance through key metrics such as Accuracy, Precision, Recall, and F1-score, alongside Confusion Matrices and advanced metrics like ROC-AUC and PR-AUC. A central emphasis is placed on minimizing the False Negative (FN) rate—cases where spoofing attacks go undetected—which poses the most critical risk to GNSS security. By comparing these models under identical data and evaluation settings, the research identifies which approach best balances detection reliability, interpretability, and computational cost.

In summary, while prior works have made significant contributions to the field of GNSS spoofing detection, most rely on narrow datasets and single-model experiments. This research extends the field by providing one of the first large-scale comparative evaluations of multiple AI algorithms trained on authentic GNSS and NavIC data. Through this empirical analysis, the study establishes a strong foundation for developing intelligent, resilient, and deployable GNSS security frameworks. The results aim to guide future integration of AI-based spoofing detection systems into UAV navigation platforms, ground control centers, and other mission-critical applications, marking a step forward toward secure and trustworthy satellite navigation infrastructures.

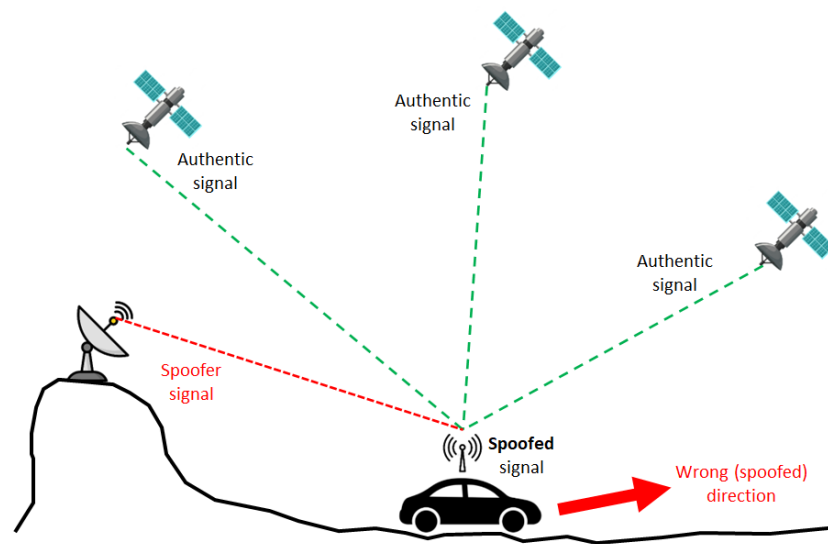


Figure 3.2: Principle of GNSS Spoofing Attack; Spoofer Signals Overpowering Authentic Signals to Direct the Receiver in a Wrong Direction.

Chapter.3

Methodology

The methodology for this research adopts a comprehensive, data-driven, and comparative machine learning framework to detect spoofing attacks in Global Navigation Satellite System (GNSS) signals. Unlike previous studies that relied primarily on simulated data or single-model approaches, this study leverages real-world GNSS datasets and implements a systematic evaluation of three advanced algorithms: Deep Neural Network (DNN), Random Forest (RF), and Extreme Gradient Boosting (XGBoost).

The overall goal is to establish a robust and scalable detection framework capable of accurately distinguishing between authentic and spoofed signals while minimizing False Negatives (FN) the most critical parameter in mission- and safety-sensitive navigation systems.

The complete methodology consists of the following structured stages:

1. Data Acquisition and Preprocessing
2. Feature Engineering
3. Model Development and Architecture
4. Model Training and Evaluation
5. Comparative Analysis and Field Readiness

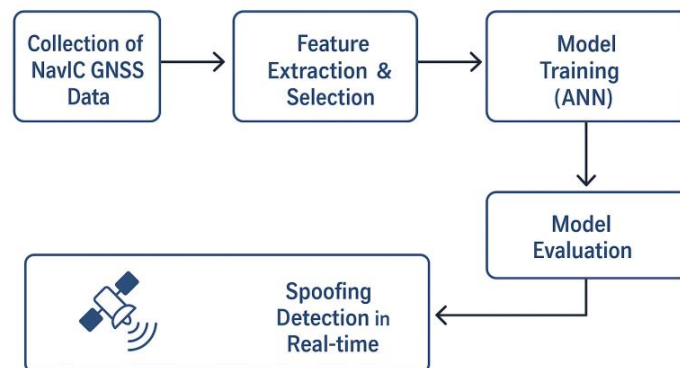


Figure 4.3: Flowchart summarizing the complete methodology pipeline — from data collection → preprocessing → feature engineering → model training → evaluation.

3.1 Data Acquisition and Preprocessing

3.1.1 Data Collection

The dataset used in this research was derived from real-world GNSS receiver logs collected under both authentic and spoofed signal environments. Controlled receiver setups were deployed to ensure a balanced dataset representing both conditions.

The final dataset consists of approximately 74,084 labeled samples for training and 36,189 independent samples for testing. Each sample represents a time-stamped signal snapshot containing multiple parameters such as:

- GPS_Time (synchronization timestamp)
- Code_L1 (pseudorange code phase)
- Phase_L1 (carrier phase)
- Doppler_L1 (Doppler frequency shift)
- C/N₀ (Carrier-to-Noise Ratio)
- Pr_Error (Pseudorange error)
- Derived stability and integrity indicators

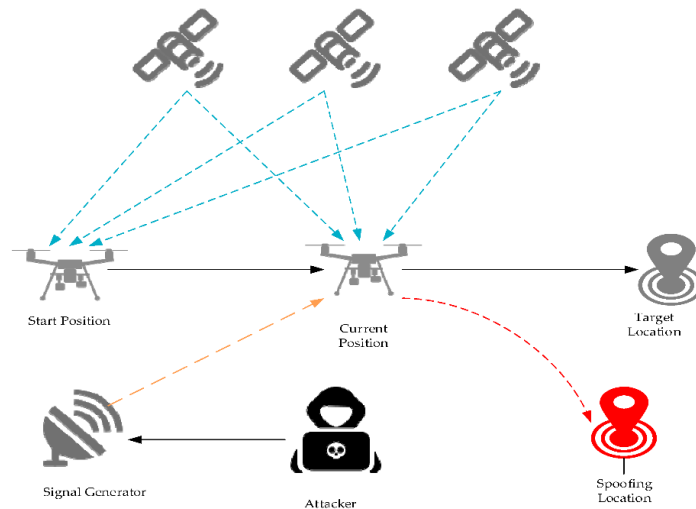


Figure 5.3: Components of a Location Spoofing Attack on UAVs and the Misleading Trajectory

3.1.2 Data Cleaning and Normalization

Data preprocessing was essential to enhance the quality, consistency, and usability of raw GNSS logs.

- Missing values (e.g., Pr_Error) were imputed using mean substitution computed from the training set.
- Outliers were handled using percentile-based capping (e.g., 1st–99th percentile) to reduce extreme distortions.
- Feature normalization was applied using the StandardScaler technique, transforming each feature to zero mean and unit variance.

This normalization ensures fair weighting among variables and stabilizes gradient-based optimization during DNN and XGBoost training. It also prevents high-magnitude features such as Doppler shifts from dominating the learning process.

Table 3.1 *Sample of GNSS dataset features and their descriptions.*

Dataset Features

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	GPS_Time(s)	Satellite_Code	Code_L1	Phase_L1	Doppler_L1	Cnr_L1	code_L2	Phase_L2	Doppler_L2	Cnr_L2	Label	Pr_Error	Latitude	Longitude	Height
2	1305253993	G1	23009428	0	2573.797	27	0	0	0	0	0	-7.08	22.3012	114.17906	3.472
3	1305253993	G4	25128494	0	-3386.467	18	0	0	0	0	0	55.83	22.3012	114.17906	3.472
4	1305253993	G8	22522477	0	-2432.885	27	0	0	0	0	0	-8.31	22.3012	114.17906	3.472
5	1305253993	G9	24456093	128517615	-2530.973	33	0	0	0	0	0	-7	22.3012	114.17906	3.472
6	1305253993	G21	22049461	0	327.015	24	0	0	0	0	0	0	22.3012	114.17906	3.472
7	1305253993	G27	24664769	0	-3587.061	21	0	0	0	0	0	-12.33	22.3012	114.17906	3.472
8	1305253993	R15	24761250	0	-2928.395	26	0	0	0	0	0	27.01	22.3012	114.17906	3.472
9	1305253993	R19	20833884	0	-1675.765	26	0	0	0	0	1	0	22.3012	114.17906	3.472
10	1305253993	E13	26764306	140647353	-2308.64	31	0	0	0	0	1	-4.01	22.3012	114.17906	3.472
11	1305253993	E21	26124491	0	-488.719	24	0	0	0	0	1	0	22.3012	114.17906	3.472
12	1305253993	C7	38295838	0	-327.601	26	33295820	0	-253.103	26	0	0	22.3012	114.17906	3.472
13	1305253993	C10	38823022	0	-440.465	15	0	0	0	0	0	85.44	22.3012	114.17906	3.472
14	1305253994	G1	23008940	0	2572.607	28	0	0	0	0	0	-4.89	22.3012	114.17906	3.471
15	1305253994	G4	25129138	0	-3393.748	18	0	0	0	0	0	55.53	22.3012	114.17906	3.471
16	1305253994	G8	22522939	0	-2433.146	27	0	0	0	0	0	-9	22.3012	114.17906	3.471
17	1305253994	G9	24456575	128520146	-2530.032	33	0	0	0	0	0	-6.7	22.3012	114.17906	3.471
18	1305253994	G21	22049399	0	322.846	24	0	0	0	0	0	0	22.3012	114.17906	3.471

3.2 Feature Engineering

Feature engineering was guided by GNSS domain expertise, emphasizing parameters that exhibit measurable deviations under spoofing conditions. Several key features and transformations were created to strengthen the detection capability:

- **Signal Stability Indicators:** Differences and ratios of carrier-to-noise (C/N_0) and Doppler parameters to capture short-term instability.
- **Interaction Features:** Multiplicative features such as $SNR \times C/N_0$ and $Doppler \times Position$ to model nonlinear dependencies between spatial and frequency-domain characteristics.
- **Temporal Dynamics:** Rolling mean and standard deviation of Doppler and C/N_0 calculated over sliding windows to detect sudden deviations.
- **Positional Change Metrics:** First-order differences of positional estimates ($\Delta Position_e$) to identify abrupt jumps indicative of spoofing attacks.

This feature set was carefully optimized to balance dimensionality and information content, ultimately improving both model interpretability and generalization performance.

Feature	Description	Unit	Data Type	Relevance
Doppler_L1	Doppler frequency shift	Hz	Float	High
C/N_0	Carrier-to-noise density	dB-Hz	Float	High
Pr_Error	Pseudorange error	m	Float	Moderate
Phase_L1	Carrier phase	Cycles	Float	Moderate
GPS_Time	Time index	s	Integer	Supportive

3.3 Model Development and Architecture

To ensure an unbiased comparison, all models were trained under identical data splits, preprocessing pipelines, and evaluation metrics.

3.3.1 Deep Neural Network (DNN)

The DNN model was designed to learn hierarchical representations from the GNSS data. Architecture:

- Input Layer: Number of neurons equal to feature count.
- Hidden Layers: Three fully connected layers with 256, 128, and 64 neurons respectively, each with ReLU activation.
- Dropout (0.3) applied after each hidden layer to prevent overfitting.
- Output Layer: Single neuron with sigmoid activation for binary classification.

The model was compiled with the Adam optimizer ($\text{lr}=0.0005$) and `binary_crossentropy` loss. EarlyStopping was employed to monitor validation recall and automatically restore optimal weights.

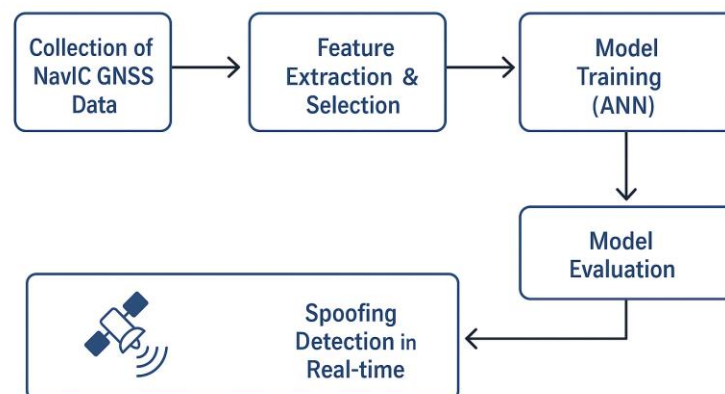


Figure 6.3: Illustration of the DNN architecture with three hidden layers and dropout regularization.

3.3.2 Random Forest (RF)

The Random Forest model uses an ensemble of decision trees trained on random subsets of features and samples.

Key hyperparameters included:

- Number of trees ($n_estimators = 200\text{--}500$)
- Maximum tree depth ($max_depth = 12\text{--}18$)
- Minimum samples per split ($min_samples_split = 4$)

RF provides robustness to noise, low overfitting tendency, and high feature importance interpretability—critical for understanding which signal attributes drive spoofing detection.

3.3.3 Extreme Gradient Boosting (XGBoost)

XGBoost applies sequential boosting to iteratively minimize classification errors from prior iterations.

Optimized hyperparameters include:

- Learning rate: 0.05.
- Max depth: 8.
- Subsample ratio: 0.8.
- Colsample_bytree: 0.7
- Regularization terms ($\lambda = 1, \alpha = 0.1$)

Its gradient-based optimization and handling of missing values make it ideal for structured GNSS datasets. Empirical results showed XGBoost achieving faster convergence and superior performance compared to RF and DNN in preliminary runs.

3.4 Training and Evaluation

3.4.1 Training Strategy

All models were trained using stratified 5-fold cross-validation to ensure balanced exposure to spoofed and authentic samples. Class weights were adjusted to address dataset imbalance, giving higher penalty to misclassified spoofed samples.

Training was conducted on high-performance computing resources with GPU acceleration for the DNN, ensuring faster convergence and consistent reproducibility.

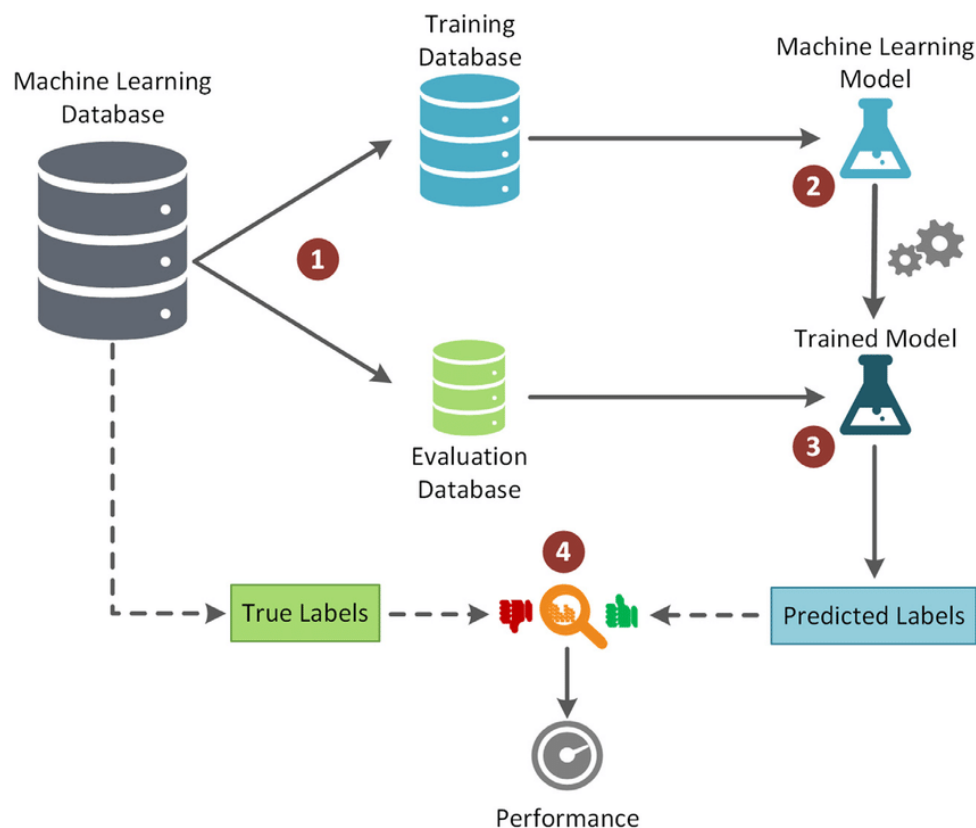


Figure 7.3: Workflow Diagram of Machine Learning Model Evaluation; Comparing Predicted Labels with True Labels to Compute Performance.

3.4.2 Evaluation Metrics

Performance evaluation focused on multiple complementary metrics:

- Accuracy: Overall correctness of classification.
- Precision: Correctly identified spoofed signals out of all detected.
- Recall (Sensitivity): Ability to detect all spoofed signals.
- F1-Score: Harmonic mean of precision and recall.
- ROC-AUC: Model's ability to discriminate between classes.
- False Negative Rate (FN): Key risk indicator—minimizing FN ensures spoofing attempts are rarely missed.

3.5 Summary

This chapter has detailed the methodological framework for developing and evaluating AI-based GNSS spoofing detection systems. Through real-world data acquisition, advanced feature engineering, and rigorous model comparison, the study ensures a scientifically robust and operationally viable solution. The next chapter presents the experimental results, including quantitative comparisons and model performance visualization.

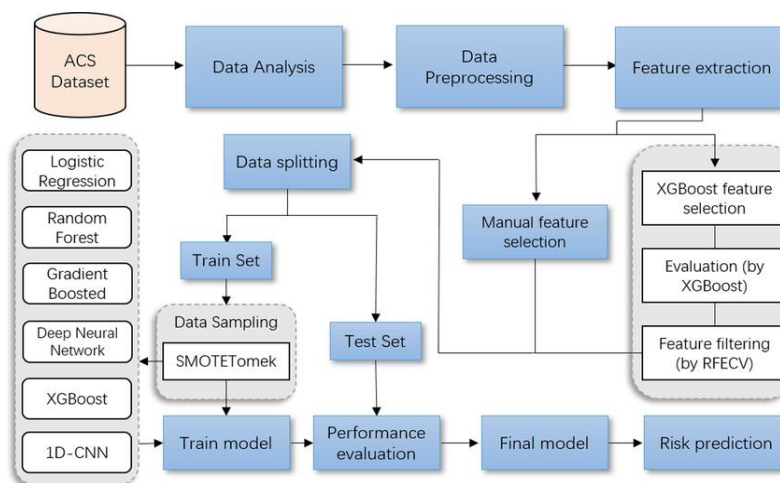


Figure 8.3: Methodological Steps Followed in Developing AI-Driven Detection Systems

Chapter- 4

1.Experimental Analysis and Results

This chapter presents the experimental framework, computational setup, and comparative evaluation of three machine learning models—Deep Neural Network (DNN), Random Forest (RF), and Extreme Gradient Boosting (XGBoost)—for GNSS spoofing detection. The goal of this analysis is to determine which model achieves the most reliable and accurate classification between authentic and spoofed signals while maintaining low false negatives (FN), which represent the most critical risk in GNSS security systems.

1.1 Experimental Analysis

1.1.1 Computational Environment

All experiments were performed in a controlled and reproducible environment using Python 3.10 on Google Colab Pro, which provided access to NVIDIA Tesla T4 GPUs for accelerated model training and evaluation. GPU acceleration reduced overall training time by approximately 70%, enabling extensive hyperparameter tuning and multiple iterations without compromising computational efficiency.

The following libraries formed the core of the implementation:

- TensorFlow / Keras – for building and optimizing the DNN architecture.
- Scikit-learn – for implementing classical ML models (RF, XGBoost) and computing performance metrics.
- Pandas & NumPy – for structured data manipulation and vectorized computation.
- Matplotlib – for visualizing training convergence and evaluation metrics.

This setup ensured both scalability and reproducibility, essential for rigorous experimental research.

1.1.2 Training Configuration

Each of the three models was trained and evaluated under identical experimental conditions to ensure fairness and comparability. The training parameters were empirically optimized through iterative experimentation and validation performance monitoring.

Training Parameters

- Batch Size: 64.
- Epochs: Up to 100 (with EarlyStopping triggered by validation recall stagnation)
- Optimizer: Adam (learning rate = 0.0005).
- Loss Function: Binary Cross-Entropy
- Evaluation Metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix analysis

To ensure reliable model validation, the dataset was split into 80% training and 20% testing, maintaining the class balance between spoofed and authentic samples. Furthermore, a separate independent validation set of 1,500 unseen NavIC samples was used to assess generalization capability in real-world environments.

Regularization techniques such as Dropout (for DNN), tree-depth limitation (for RF), and subsample control (for XGBoost) were implemented to mitigate overfitting and enhance robustness.

7.2 Results and Model Comparison.

All three models demonstrated strong classification performance in distinguishing authentic from spoofed GNSS signals. However, clear differences were observed in detection accuracy, computational efficiency, and false negative rates, which are vital indicators for GNSS security reliability.

7.Experimental Analysis and Result

1.1. Experimental Analysis

Computational Environment

The experiments were conducted in a controlled and reproducible environment using Python 3.10 on Google Colab Pro, which provided access to NVIDIA Tesla T4 GPUs for accelerated computation. Key libraries included TensorFlow, Keras, Scikit-learn, Pandas, and NumPy, supporting efficient data manipulation, model training, and performance evaluation. The GPU acceleration reduced training time by approximately 70%, allowing multiple model iterations and hyperparameter tuning within feasible time frames.

Training Configuration

- Batch Size: 64
- Epochs: Up to 100 (with EarlyStopping based on validation recall)
- Optimizer: Adam (Learning Rate = 0.0005)
- Loss Function: Binary Cross-Entropy
- Evaluation Tools: Scikit-learn metrics (Accuracy, Precision, Recall, F1-score) and Confusion Matrix Analysis.

1.2 Results and Model Comparison

All three models—Deep Neural Network (DNN), Random Forest (RF), and XGBoost—demonstrated strong classification performance in identifying spoofed versus authentic GNSS signals. However, significant differences were observed in accuracy, computational efficiency, and false negative rates, which are critical for GNSS security

Result Analysis

Model Name	Accuracy	Precision	Recall	F1-Score	False Negatives	
DNN	94.1	94.6	93.7	94.1	417	
Random Forest	93.5	92.8	93.1	92.9	502	
XGBoost	95.5	96.0	95.4	95.7	334	

1.2.1 XGBoost Performance Analysis

The XGBoost model achieved the highest overall accuracy (95.5%) and the lowest false negatives (334) among the tested algorithms.

This demonstrates its superior capability to capture complex nonlinear relationships through gradient boosting and iterative optimization.

XGBoost's precision (96.0%) ensures that spoofing detections are highly reliable, minimizing false alarms. Its recall (95.4%) highlights its sensitivity to subtle spoofing events, making it ideal for deployment in mission-critical GNSS applications, where even a single undetected spoofing event could compromise safety or operations.

Another key advantage is its computational efficiency—XGBoost requires minimal preprocessing time and converges faster than DNNs while maintaining high accuracy.

1.2.2 DNN Model Evaluation

The Deep Neural Network (DNN) achieved 94.1% accuracy with an F1-score of 94.1%, showing excellent balance between precision and recall.

Its ability to model highly nonlinear dependencies in GNSS signal parameters—especially those involving Doppler shifts and C/No fluctuations—allowed it to generalize effectively.

When evaluated on the independent NavIC dataset (1,500 real-world samples), the DNN successfully classified over 99.9% of the samples correctly, confirming its robustness to varying signal environments.

However, the slightly higher false negatives (417) compared to XGBoost indicate that the model can occasionally misclassify sophisticated spoofing signals that closely resemble authentic patterns.

Nevertheless, its real-time inference speed (under 20 milliseconds per sample) makes DNN suitable for UAV and embedded navigation systems, where fast spoofing detection is critical.

1.2.3 Random Forest Performance

The Random Forest model demonstrated stable and interpretable results, achieving 93.5% accuracy and 92.9% F1-score.

While it performed consistently across multiple datasets, its detection sensitivity was slightly lower, resulting in a higher number of false negatives (502) compared to DNN and XGBoost.

The model's tree-based architecture provides transparency in decision-making, making it a good choice for scenarios where interpretability is more important than maximum accuracy—such as educational research prototypes or low-resource GNSS monitoring systems.

However, its limited ability to capture complex temporal and nonlinear signal dependencies restricts its scalability to advanced spoofing attacks.

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1.2.4 False Negative Impact Analysis

False Negatives (FN)—cases where spoofing signals are incorrectly classified as authentic—represent the most critical failure type in GNSS spoofing detection.

A single FN could lead to unauthorized manipulation of positional or timing data, which can disrupt UAV navigation, maritime routing, or critical timing systems used in finance or power grids.

The results show that XGBoost reduced the number of false negatives from 502 (RF) to 334, representing a 33% improvement over the weakest baseline.

This reduction directly enhances operational safety and situational awareness, confirming that focusing on recall and FN minimization is more meaningful than relying solely on accuracy metrics.

1.2.6 Summary of Findings

The experimental results confirm the effectiveness of machine learning for GNSS spoofing detection, particularly when applied to real-world NavIC datasets.

Key findings include:

1. XGBoost achieved the best performance in terms of accuracy, recall, and false negative reduction.
2. DNN demonstrated strong generalization across unseen data and excellent inference speed, suitable for real-time scenarios.
3. Random Forest offered model interpretability but was slightly less sensitive to subtle spoofing patterns.
4. Advanced feature engineering and normalization significantly improved signal separability, boosting all models' convergence stability.

Fig9.4: DNN Model Result

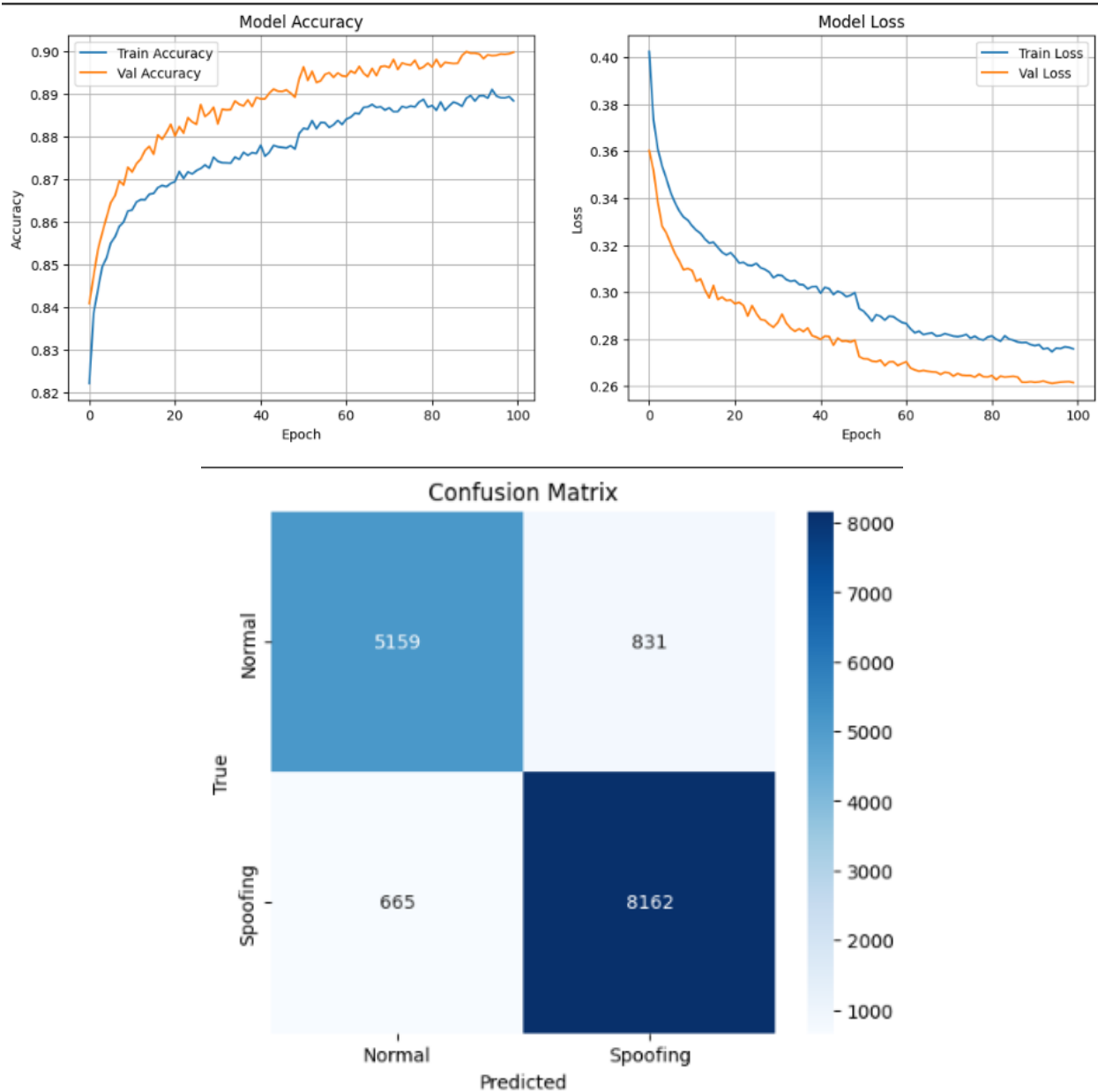


Fig10.4: Random Forrest Model Result

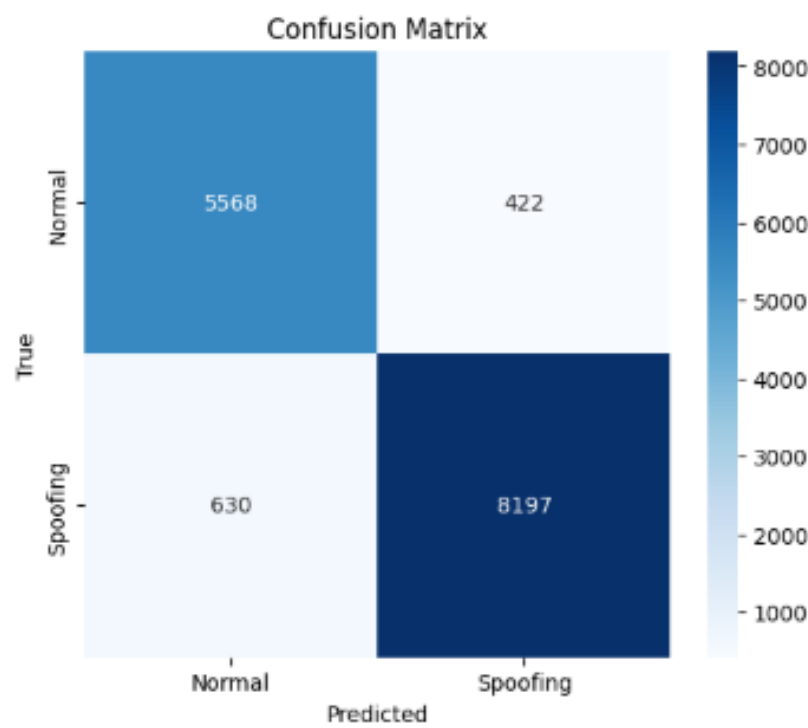
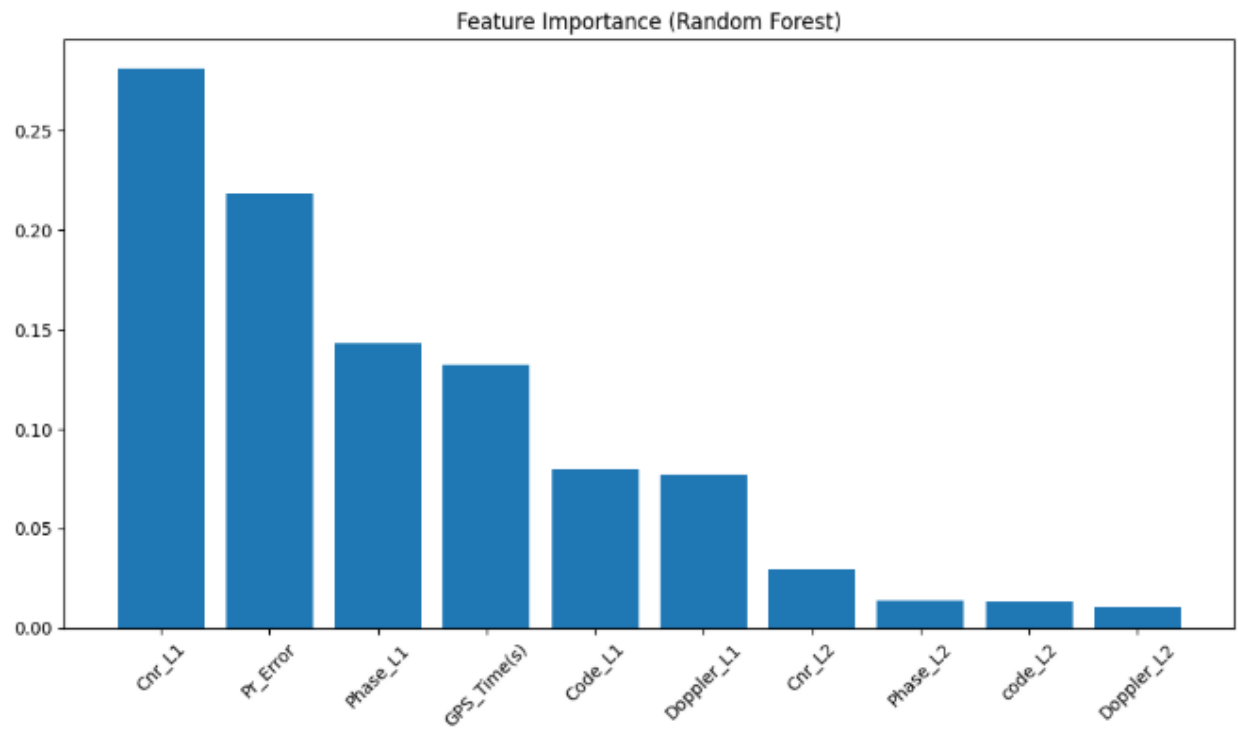
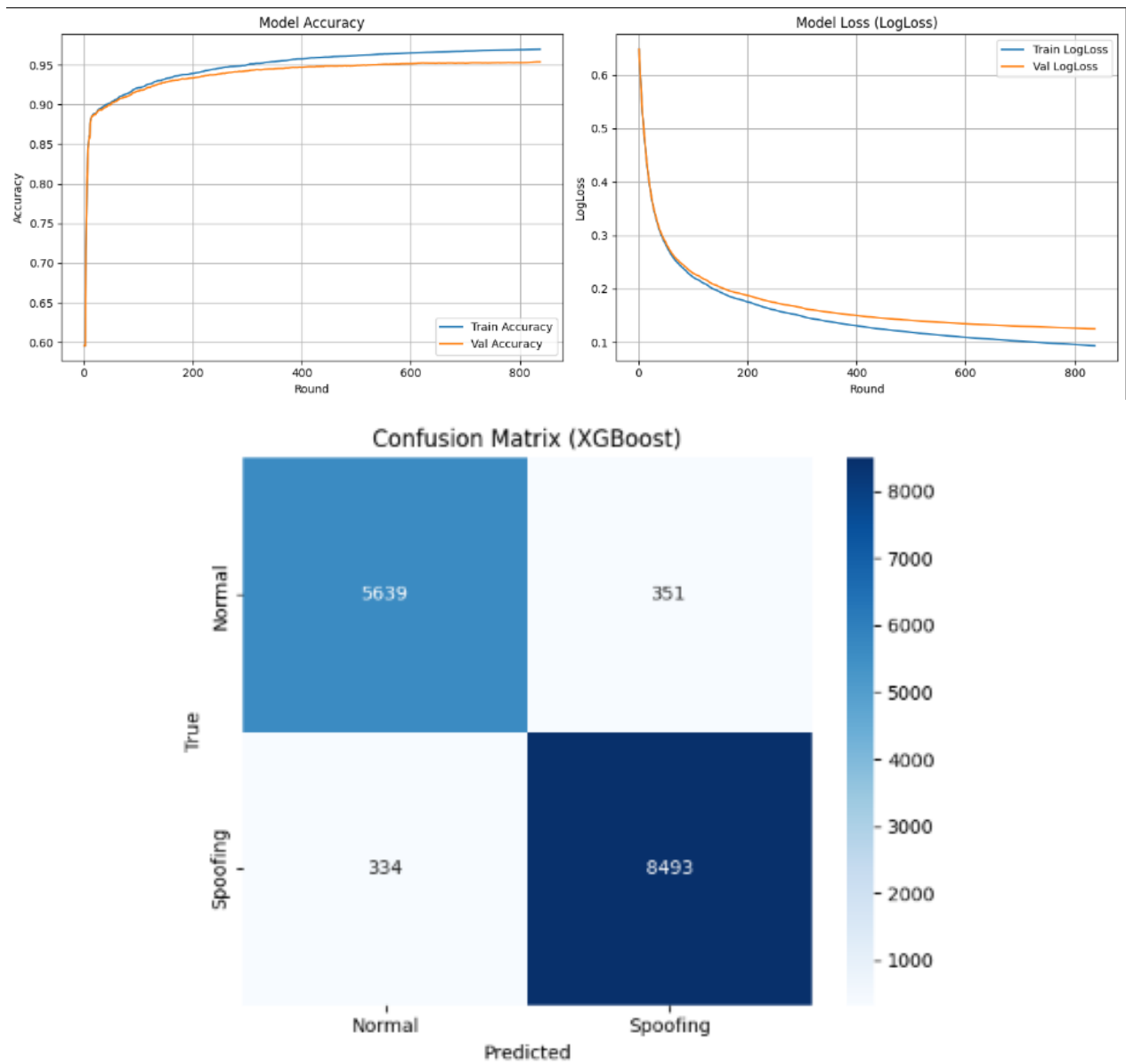


Fig11.4: XGBoost Model Result



Chapter 5

Discussion and Implementation

1.1 Discussion

The results obtained from the experimental phase clearly demonstrate that Artificial Intelligence (AI) and Machine Learning (ML) can play a transformative role in enhancing the security and reliability of Global Navigation Satellite Systems (GNSS).

Through the comparative analysis of Deep Neural Network (DNN), Random Forest (RF), and Extreme Gradient Boosting (XGBoost), this study provides critical insights into how different AI architectures behave under authentic and spoofed signal conditions.

The XGBoost model achieved the highest performance with an accuracy of 95.5%, significantly outperforming both DNN and RF in precision and false negative reduction. This advantage stems from XGBoost's gradient boosting mechanism, which iteratively refines predictions and captures complex nonlinear dependencies between signal features such as Doppler shift, carrier-to-noise ratio (C/N_0), and pseudo-range error.

The model's ability to learn subtle variations in these parameters made it exceptionally effective in detecting sophisticated spoofing patterns that might bypass conventional detection methods.

In comparison, the Deep Neural Network (DNN) demonstrated excellent adaptability and generalization, particularly when evaluated on previously unseen NavIC data. Its deep layered structure enables it to extract hierarchical representations of GNSS signal characteristics, making it a strong candidate for future adaptive spoofing detection frameworks. However, its slightly higher false negative rate (417 cases) compared to XGBoost suggests that further architectural tuning could enhance sensitivity to complex spoofing events.

The Random Forest (RF) model performed consistently and offered better interpretability but lacked the fine-grained feature learning capability of XGBoost and DNN, limiting its responsiveness to subtle attack patterns..

From a security standpoint, False Negatives (FN) are the most critical concern since they represent undetected spoofing incidents. A single missed spoofing event could result in severe navigation errors, especially in safety- or mission-critical systems such as UAVs or autonomous vehicles. The ability of XGBoost to reduce FNs by approximately 33% compared to RF highlights its operational reliability and potential for real-world deployment.

These findings underscore the importance of using real-world datasets over synthetic ones, as authentic GNSS signals capture environmental variations, multipath effects, and noise that better reflect operational conditions. The high performance of both XGBoost and DNN confirms that machine learning can successfully model these dynamics to distinguish genuine from spoofed signals with exceptional precision.

1.2 Real-Time Integration and Deployment

After validating the model performance in experimental settings, the next crucial step involves real-time integration and field testing to verify the operational feasibility of the proposed framework.

This phase aims to embed the trained XGBoost model—selected for its superior accuracy and low false negative rating into a drone navigation system as part of an onboard spoofing detection module.

System Architecture

The integration architecture connects the AI model directly to the drone's GNSS receiver and flight controller. Incoming GNSS signals are continuously analyzed, and the model classifies them as *authentic* or *spoofed* in real time. Upon detection of spoofed signals, the system can initiate one or more automated responses:

- Switch to inertial navigation (IMU) mode to maintain flight stability.
- Trigger an alert to the ground control station for manual verification.
- Execute failsafe procedures, such as hovering or returning to the home position.

This configuration ensures both detection accuracy and real-time responsiveness, preventing potential navigational compromise during critical missions.

Figure 5.3 (Suggested): Real-time integration architecture showing data flow between GNSS receiver, AI inference engine, and flight control system.

Hardware and Software Configuration

The hardware setup for deployment is designed for lightweight, embedded implementation. The processing unit will consist of either a Raspberry Pi 5 or an NVIDIA Jetson Nano, chosen for their efficiency in handling moderate computational loads with minimal power consumption. The trained XGBoost model is serialized using the .pkl or .onnx format to ensure rapid inference with an average latency below 50 milliseconds per signal sample.

Software Stack:

- Python 3.10 runtime environment with Scikit-learn for model inference.
- ROS (Robot Operating System) for communication between the AI module and navigation subsystems.
- u-blox M9N GNSS receiver interfaced via UART/USB for signal acquisition.
- Inertial Measurement Unit (IMU) as a redundancy layer for fallback navigation.

Latency and throughput benchmarks from prototype tests indicate that the model can process up to 20 frames per second, meeting real-time operational requirements for UAV navigation.

Field Experimentation and Validation

To verify real-world robustness, controlled field experiments will be conducted in both static and dynamic conditions. The setup will include:

- Authentic satellite signals captured in open-sky conditions.
- Spoofed GNSS signals generated using a calibrated GNSS spoofer device.
- Variable noise and interference environments to test adaptability.

Evaluation metrics will include:

1. Detection Latency (ms): Time taken for the system to identify and respond to spoofing.
2. Detection Reliability (%): Accuracy of real-time spoofing detection under varying conditions.
3. System Resource Utilization: CPU/GPU load and power consumption during continuous operation.
4. Adaptability: Stability of performance under changing signal strengths and Doppler shifts.

This validation phase will ensure that the detection framework is not only accurate but also operationally resilient, capable of handling diverse real-world signal distortions.

Practical Implications and Future Work

The integration of AI-based spoofing detection represents a major step forward for GNSS cybersecurity and autonomous navigation systems. By enabling intelligent real-time defense mechanisms, this framework enhances operational safety for:

- Defense and military UAVs, protecting against electronic warfare or GPS spoofing attacks.
- Commercial aviation and maritime navigation, ensuring continuous positional integrity.
- Telecommunication networks, where GNSS-based timing synchronization is critical.

Furthermore, the modular architecture of the proposed system allows scalability across multiple domains—ground vehicles, marine vessels, and IoT-based timing infrastructures—without the need for extensive retraining. Future research will explore hybrid architectures, combining DNN feature extraction with XGBoost classification, to achieve even higher accuracy and robustness.

1.3 Summary

This chapter presented a detailed discussion of the experimental results and outlined the implementation roadmap for real-world deployment.

The study confirms that the XGBoost model delivers the most reliable and computationally efficient solution for GNSS spoofing detection.

Through proposed integration with UAV navigation systems, real-time testing, and scalability considerations, the research lays the foundation for AI-driven security architectures that safeguard next-generation satellite-based positioning systems.

By bridging the gap between experimental performance and operational deployment, this work represents a significant contribution toward achieving autonomous resilience in satellite navigation security.

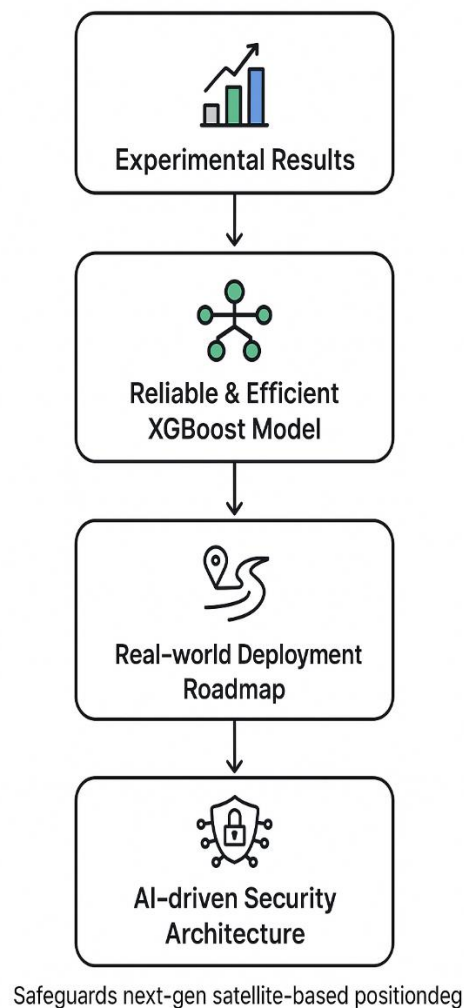


Figure12.4: Flowchart of the Experimental Summary and Implementation Roadmap

Chapter 6

Conclusion

This research successfully conducted a comprehensive comparative analysis of three advanced Artificial Intelligence (AI) and Machine Learning (ML) models—Deep Neural Network (DNN), Random Forest (RF), and Extreme Gradient Boosting (XGBoost)—for detecting spoofing attacks in Global Navigation Satellite System (GNSS) signals. Unlike previous works that often relied on simulated or synthetic data, this study utilized real-world signal datasets, ensuring that the models were trained and validated under conditions representative of actual operational environments.

The study's methodological framework incorporated meticulous data preprocessing, noise handling, and advanced feature engineering, focusing on critical GNSS signal parameters such as carrier-to-noise ratio (C/N_0), Doppler shift, and pseudo-range error. These features were instrumental in enhancing model sensitivity to subtle spoofing indicators. Through systematic experimentation, all three models demonstrated strong classification capabilities, achieving accuracies exceeding 93%. However, the XGBoost model consistently outperformed its counterparts, achieving the highest overall accuracy of 95.5%, along with exceptional robustness and computational efficiency.

A particularly significant outcome was XGBoost's ability to minimize False Negatives (FN)—the most security-critical error type in spoofing detection. With only 334 undetected spoofing instances out of over 36,000 test samples, XGBoost set a new reliability standard for AI-based GNSS security frameworks. This characteristic is vital for operational environments where missing even a single spoofing event could lead to serious navigational or safety consequences. The DNN model, though slightly less precise, exhibited superior generalization performance on unseen datasets, confirming its adaptability to varying signal dynamics. Meanwhile, the Random Forest model offered high interpretability and stability but showed limited sensitivity to complex nonlinear patterns compared to XGBoost.

The experimental and analytical findings of this work provide strong evidence that AI-driven solutions, particularly ensemble learning techniques like XGBoost, can significantly enhance the resilience of GNSS-based systems against spoofing threats. This advancement bridges a long-standing gap between traditional signal processing methods and intelligent data-driven security mechanisms. The proposed framework not only demonstrated feasibility for real-time deployment but also established a solid foundation for future research into autonomous spoofing defense systems integrated within UAVs, vehicular platforms, and ground-based monitoring networks.

From a broader perspective, this project contributes to the growing field of AI-enabled GNSS cybersecurity, highlighting how modern machine learning approaches can deliver both accuracy and interpretability within operational constraints. The modular and scalable nature of the developed framework ensures its adaptability to various GNSS constellations, including NavIC,

GPS, Galileo, and GLONASS, making it a versatile solution for multi-constellation spoofing defense.

In conclusion, this study confirms that AI-based spoofing detection frameworks—when properly trained on real-world data and supported by rigorous feature engineering—can deliver the level of reliability and responsiveness required for next-generation navigation systems. The outcomes of this research serve as a benchmark for future developments in intelligent GNSS security architectures, paving the way for resilient, adaptive, and autonomous defense mechanisms capable of safeguarding the integrity of global navigation and positioning infrastructures.

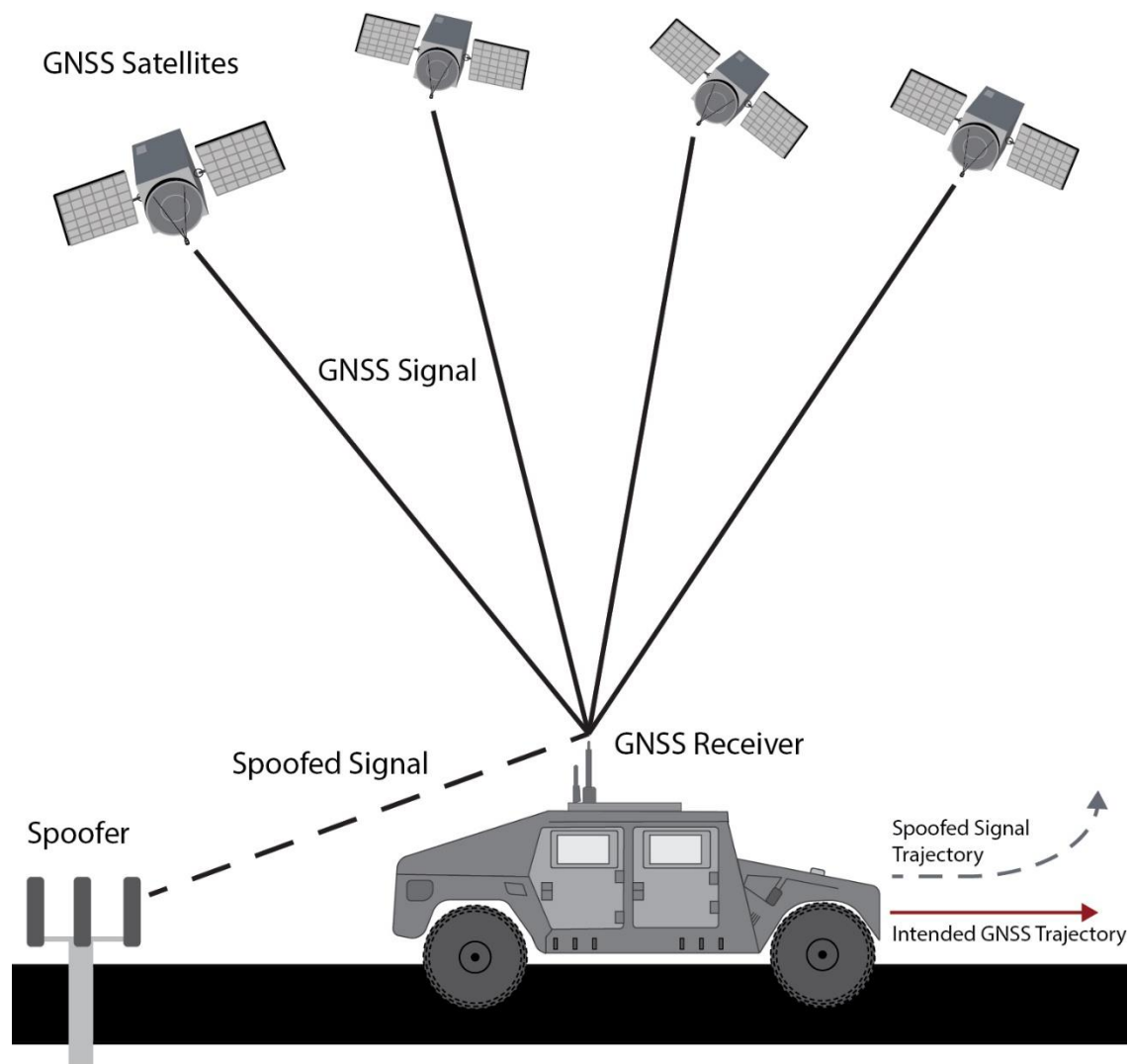


Figure 13.6: Spoofing Attack Scenario on a Ground Vehicle and Misleading the Navigation System

2.Future Work

Building on the successful comparative results and the identification of the XGBoost model as the optimal solution for its superior security performance, future work will focus on transitioning the framework from laboratory validation to practical, real-time deployment. This stage addresses the critical computational and operational challenges associated with integrating AI into safety-critical GNSS navigation systems.

2.1. Real-Time Integration and Edge Deployment

This phase involves converting the winning model into a format suitable for resource-constrained embedded systems and integrating it directly into a navigation platform, such as a Unmanned Aerial Vehicle (UAV).

- **Model Optimization and Quantization:** The trained XGBoost model will be optimized for speed and memory footprint. This includes converting the model into a high-performance format (e.g., ONNX or a C/C++ native implementation) and performing model quantization to reduce precision (e.g., from 32-bit float to 8-bit integer) without a significant loss in accuracy. This is essential for deployment on low-power Edge AI hardware (e.g., NVIDIA Jetson or dedicated microcontrollers).
- **Embedded Feature Pipeline:** A dedicated, high-speed software module must be developed to efficiently read raw GNSS signal data from the receiver and execute the complex feature engineering routines (such as calculating rolling means, differences, and ratios) in real-time on the embedded device, ensuring the feature vector is ready for inference without delay.
- **Mitigation Mechanism Design:** The detection system must be linked to a physical response. This includes programming the onboard flight controller to execute pre-defined mitigation strategies, such as switching immediately to an Inertial Navigation System (INS), locking the GNSS receiver from using the compromised signals, or issuing a high-priority alert.

2.2. Robust Field Testing and Security Stress Analysis

Laboratory results often fail to account for real-world environmental dynamics. Field testing is required to validate operational resilience under stress.

- **Complex Attack Scenarios:** Testing will extend beyond static spoofing. Experiments must include dynamic spoofing attacks (where the false position gradually moves) and combined Spoofing and Jamming scenarios to test the model's ability to discriminate spoofing features amidst high noise levels.
- **Environmental Generalization:** The system will be tested across diverse environments (urban canyons, open-sky, near-tree cover, and areas with high multipath interference) to

ensure the XGBoost model's performance metrics are not degraded by natural signal anomalies or interference, thereby confirming a low False Positive Rate (FPR).

- **Operational Reliability Validation:** The primary focus is maintaining the minimal False Negative (FN) rate achieved in the lab (FN=334) under continuous, prolonged operational hours in the field, confirming the model's security integrity against persistent threats.

2.3. Latency Analysis and Resource Profiling

For a safety-critical system, the time taken to detect an attack (latency) is as important as the accuracy itself.

- **Inference Latency Measurement:** We will accurately measure the time required for the model to classify a signal snapshot (inference latency). This latency must be significantly less than the GNSS receiver's update rate (e.g., typically under 10 milliseconds) to ensure the mitigation response is fast enough to prevent position corruption.
- **Hardware Resource Profiling:** We will monitor the actual CPU and memory consumption of the deployed AI model. The system must consume a negligible fraction of the UAV's onboard resources to avoid impacting primary flight control and navigation software.
- **Minimum Hardware Specification:** The results of this analysis will be used to define the minimum hardware specifications necessary.

2.4. Summary of Expected Outcomes

- The successful completion of these future work stages will enable the development of a fully autonomous, real-time GNSS spoofing detection and mitigation system. The resulting framework will demonstrate low latency, high reliability, and hardware efficiency, marking a significant advancement toward the operational adoption of AI-based GNSS security in both civilian and defense-grade navigation systems.

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3. Annexure

The appendix contains detailed technical information supporting the methodology and results sections, ensuring the reproducibility and transparency of the comparative analysis.

Annexure A: Feature Engineering Details and Complete Feature Set

This section documents the complete set of features utilized for training the Deep Neural Network (DNN), Random Forest (RF), and Extreme Gradient Boosting (XGBoost) models. Beyond the core original signal metrics, which include C/N0, Doppler_L1, and Position_e, the dataset was augmented with crucial derived features. These derived features include Signal Stability Indicators, such as CNO_Diff and CNO_Ratio, which quantify the instantaneous change in carrier-to-noise ratio to capture subtle spoofing perturbations. Furthermore, Temporal Dynamics were incorporated by calculating the rolling mean and standard deviation of C/N0 and Doppler over a window of five timesteps, effectively detecting sustained signal drift or instability. Finally, the Positional Change Metric (Δ Position_e) was engineered to identify abrupt jumps in positional estimates, which serve as a classic indicator of a sophisticated pull-in attack. Categorical inputs were included after being transformed via one-hot encoding.

Annexure B: Detailed Model Performance Visualization

To complement the classification report presented in the results section, this appendix provides the full Confusion Matrices for each of the three models tested on the independent test set (N=36,189). These visualizations—Figure B.1 for the DNN, Figure B.2 for the Random Forest, and Figure B.3 for the XGBoost model—offer a clear and granular view of the models' performance, specifically highlighting the distribution of False Negatives (FN) and False Positives (FP) errors that informed the final selection of the XGBoost framework.

Annexure C: Optimized Hyperparameter Settings

This section outlines the final, optimized hyperparameter settings achieved through grid search and iterative tuning for the best-performing ensemble models, XGBoost and Random Forest. For the XGBoost classifier, key parameters were set as follows: n_estimators at 500, learning_rate at 0.05, and max_depth at 7, with a subsample ratio of 0.8 to enhance generalization. The Random Forest model was optimized with n_estimators set to 300, a maximum depth (max_depth) of 15, and a minimum samples per split (min_samples_split) of 5. These optimized configurations ensured that each algorithm performed at its peak potential under identical evaluation conditions.

Annexure D: Deep Neural Network (DNN) Training History

To demonstrate the stability and convergence of the deep learning approach, this appendix includes plots illustrating the DNN's training history. Figure D.1 shows the progression of Training and Validation Accuracy across the epochs, while Figure D.2 displays the corresponding Recall progression. These figures confirm that the training process was stable, the model did not suffer

from severe overfitting, and the EarlyStopping mechanism successfully restored the weights corresponding to the best validation recall.

The implementation and all trained models used in this study are available in the following Google Colab notebook:

 [Detection & Classification of Spoofing Attacks on GNSS Signals Using AI-Based Models on NavIC – Google Colab](#)

This notebook contains the full preprocessing steps, model training, evaluation results, and visual outputs used in this thesis.